
DP-ShiftBench: How Distribution Shift Governs the Value of Pretraining for Private Learning

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1 Introduction

Pretraining on public data followed by finetuning with DP-SGD on private data has become the dominant paradigm for private learning, yielding large accuracy gains over training from scratch Li et al. [2022], Ganesh et al. [2023], De et al. [2022]. However, in practice the public and private distributions rarely match—e.g., pretraining on web images but finetuning on medical scans. The effect of this distribution shift on the privacy-utility tradeoff remains poorly understood: does shift add to the privacy cost independently Bassily et al. [2023], amplify it multiplicatively, or is there a critical threshold beyond which public data stops helping Setlur et al. [2025]?

We propose **DP-ShiftBench**, a controlled benchmark where distribution shift is a *continuous, tunable parameter*. It features two tracks—images (SVHN→MNIST) and language (controlled Markov chains)—each evaluated across $\varepsilon \in \{0.5, 1, 2, 4, 8, \infty\}$ with $\delta = 10^{-5}$. Our contributions: (1) a reusable benchmark with fine-grained shift control, (2) an evaluation of which divergence metrics (FID, KL, MAUVE Pillutla et al. [2023]) best predict DP degradation under shift, and (3) an empirical test of additive vs. multiplicative vs. threshold interaction models.

2 Proposed Approach

In both tracks, we pretrain on public data and finetune with DP-SGD Abadi et al. [2016] on private data, sweeping $\varepsilon \in \{0.5, 1, 2, 4, 8, \infty\}$ ($\delta = 10^{-5}$). For each (pretrain, finetune) pair we compute divergence metrics and correlate them with DP finetuning accuracy.

2.1 Image Track: SVHN → MNIST

We pretrain on SVHN Netzer et al. [2011] (32×32 color digit images) and finetune with DP-SGD on MNIST. To handle the format mismatch, MNIST images are resized to 32×32 and replicated to 3 channels. We use a SmallCNN ($\sim 500K$ params) with GroupNorm (required for Opacus compatibility). Table 1 summarizes the experiments: we control shift via class-subset matching (pretrain on digits {0–4} vs. {5–9}) and pretraining objective (supervised vs. autoencoder). For each pair, we compute FID and proxy $\mathcal{A}$ -distance Ben-David et al. [2010] to quantify the shift magnitude.

Experiment	Pretrain	Finetune (DP)	Shift Type
Baseline	None	MNIST	No pretrain
Full shift	SVHN (full)	MNIST	Domain gap
Matched	SVHN {0–4}	MNIST {0–4}	Domain only
Mismatched	SVHN {5–9}	MNIST {0–4}	Domain + label
Unsupervised	SVHN (autoenc.)	MNIST	Task gap

Table 1: Image track experiments. Each run across $\varepsilon \in \{0.5, 1, 2, 4, 8, \infty\}$.

2.2 Language Track: Controlled Markov Chains

2.3 Analysis: Divergence Metrics and Interaction Hypotheses

3 Relevant Literature

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